

Supplementary Material: Positive-Class Weighting Induces Systematic Probability Distortion in Deep Learning Models for Chest Radiography under Deidentified-Date Cohort Shift

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Section S1: Mathematical Derivation of Bayes-Optimal Logit Shift under Weighted Binary Cross-Entropy

In this section, we provide the complete formal derivation demonstrating that positive-class weighting in binary cross-entropy (BCE) mathematically induces an additive constant logit shift of $+\ln w$ in the population Bayes-optimal classifier.

Let $X \in \mathcal{X}$ denote the input feature representation (e.g., chest radiograph), and let $Y \in \{0, 1\}$ represent the binary pathology label. Let $\eta(x) = P(Y = 1 \mid X = x)$ denote the true posterior probability under the data-generating distribution. A neural network outputs a scalar logit $z(x) \in \mathbb{R}$, yielding predicted probability $\hat{p}(x) = \sigma(z(x)) = \frac{1}{1+e^{-z(x)}}$.

The positive-weighted binary cross-entropy loss with positive scalar weight $w > 0$ for a single instance (x, y) is defined as:

$$\mathcal{L}_{\text{pos}}(y, \hat{p}(x)) = -w \cdot y \ln(\hat{p}(x)) - (1 - y) \ln(1 - \hat{p}(x)) \quad (1)$$

Under population risk minimization, the optimal model minimizes the conditional expected risk at each point x :

$$\begin{aligned} R(z(x)) &= \mathbb{E}_{Y|X=x}[\mathcal{L}_{\text{pos}}(Y, \sigma(z(x)))] \\ &= -w \cdot \eta(x) \ln(\sigma(z(x))) - (1 - \eta(x)) \ln(1 - \sigma(z(x))) \end{aligned} \quad (2)$$

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Recall the standard identity for the logistic sigmoid derivative with respect to its logit z :

$$\frac{d}{dz}\sigma(z) = \sigma(z)(1 - \sigma(z)) \quad (3)$$

Differentiating the conditional expected risk with respect to logit $z(x)$:

$$\begin{aligned} \frac{dR}{dz} &= -w \cdot \eta(x) \frac{\sigma(z)(1 - \sigma(z))}{\sigma(z)} - (1 - \eta(x)) \frac{-\sigma(z)(1 - \sigma(z))}{1 - \sigma(z)} \\ &= -w \cdot \eta(x)(1 - \sigma(z)) + (1 - \eta(x))\sigma(z) \\ &= \sigma(z) [w \cdot \eta(x) + (1 - \eta(x))] - w \cdot \eta(x) \end{aligned} \quad (4)$$

Setting the first-order condition $\frac{dR}{dz} = 0$ to find the optimal predicted probability $\hat{p}^*(x) = \sigma(z^*(x))$:

$$\sigma(z^*(x)) [w \cdot \eta(x) + 1 - \eta(x)] = w \cdot \eta(x) \quad (5)$$

Solving directly for $\hat{p}^*(x)$:

$$\hat{p}^*(x) = \frac{w \cdot \eta(x)}{w \cdot \eta(x) + (1 - \eta(x))} \quad (6)$$

Converting optimal probability $\hat{p}^*(x)$ into the Bayes-optimal logit representation $z^*(x) = \text{logit}(\hat{p}^*(x))$:

$$\begin{aligned} z^*(x) &= \ln \left(\frac{\hat{p}^*(x)}{1 - \hat{p}^*(x)} \right) \\ &= \ln \left(\frac{\frac{w \cdot \eta(x)}{w \cdot \eta(x) + (1 - \eta(x))}}{1 - \frac{w \cdot \eta(x)}{w \cdot \eta(x) + (1 - \eta(x))}} \right) \\ &= \ln \left(\frac{w \cdot \eta(x)}{1 - \eta(x)} \right) \\ &= \ln \left(\frac{\eta(x)}{1 - \eta(x)} \right) + \ln(w) \end{aligned} \quad (7)$$

Let $z_{\text{true}}(x) = \text{logit}(\eta(x)) = \ln \left(\frac{P(Y=1|X=x)}{P(Y=0|X=x)} \right)$ denote the true population logit-odds. We obtain the exact closed-form relation:

$$z^*(x) = z_{\text{true}}(x) + \ln(w) \quad (8)$$

Equation (8) proves that positive loss weighting acts as an exact additive translation of the population Bayes-optimal logit by $+\ln w$. When $w > 1$ (standard practice to combat acute class imbalance), uncalibrated predictions suffer an artificial positive logit inflation, which systematically distorts posterior probability calibration into acute over-confidence. Crucially, subtracting the constant offset $\ln w$ ($z \mapsto z - \ln w$) recovers the undistorted log-odds $z_{\text{true}}(x)$ under ideal asymptotic conditions, establishing the theoretical foundation for the $O(1)$ zero-retraining analytic recalibration evaluated in the main manuscript.

Section S2: Comprehensive Cross-Institutional Factorial Matrix

Table S1 provides the complete $2 \times 2 \times 4$ cross-institutional factorial matrix on the official MIMIC-CXR-JPG test cohort ($N = 3,403$ frontal radiographs from 3,041 diagnostic studies across $N = 289$ unique patients). Both unweighted and positive-weighted binary cross-entropy (BCE) models are evaluated across DenseNet-121 and ResNet-50 architectures for all four thoracic abnormalities. For positive-weighted models, paired zero-parameter analytic logit recalibration ($z \mapsto z - \log w$) is evaluated alongside patient-clustered 1,000-replicate bootstrap 95% confidence intervals.

Table S1: Supplementary Table S1: Complete 16-Row Cross-Institutional External Validation Matrix on the Official MIMIC-CXR-JPG Frontal Test Cohort ($N = 3,403$ Radiographs, 3,041 Studies, 289 Unique Patients). Evaluated on 3-Seed Probability Ensembles with Image Prevalence (π), Empirical Null Reference Baselines ($\text{Brier}_{\text{null}} = \pi(1 - \pi)$), Patient-Clustered Bootstrap 95% Confidence Intervals, Brier Skill Scores ($\text{BSS} = 1 - \text{Brier}/\text{Brier}_{\text{null}}$), and Paired Analytic Recalibration ($\Delta\text{Brier} = \text{Brier}_{\text{corr}} - \text{Brier}_{\text{raw}}$).

Architecture	Objective	Pathology	Prevalence (π)	$\text{Brier}_{\text{null}}$	AUROC [95% CI]	Brier (Raw $\rightarrow$ Corr)	ΔBrier [95% CI]	BSS (Raw $\rightarrow$ Corr)
DenseNet-121	Unweighted BCE	Pleural Effusion	32.18%	0.2182	0.688 [0.656, 0.717]	0.2488	–	-0.140
DenseNet-121	Unweighted BCE	Cardiomegaly	26.33%	0.1940	0.523 [0.491, 0.556]	0.2375	–	-0.224
DenseNet-121	Unweighted BCE	Pneumonia	10.05%	0.0904	0.607 [0.560, 0.650]	0.0986	–	-0.091
DenseNet-121	Unweighted BCE	Edema	21.33%	0.1678	0.562 [0.524, 0.596]	0.2124	–	-0.266
DenseNet-121	Pos-Weighted BCE	Pleural Effusion	32.18%	0.2182	0.668 [0.636, 0.700]	0.2536 $\rightarrow$ 0.2451	-0.0084 [-0.0401, +0.0240]	-0.162 $\rightarrow$ -0.123
DenseNet-121	Pos-Weighted BCE	Cardiomegaly	26.33%	0.1940	0.525 [0.492, 0.558]	0.2907 $\rightarrow$ 0.2328	-0.0579 [-0.0790, -0.0349]	-0.499 $\rightarrow$ -0.200
DenseNet-121	Pos-Weighted BCE	Pneumonia	10.05%	0.0904	0.598 [0.553, 0.639]	0.1078 $\rightarrow$ 0.0983	-0.0095 [-0.0170, -0.0016]	-0.192 $\rightarrow$ -0.087
DenseNet-121	Pos-Weighted BCE	Edema	21.33%	0.1678	0.581 [0.549, 0.611]	0.1914 $\rightarrow$ 0.2131	+0.0217 [+0.0118, +0.0307]	-0.141 $\rightarrow$ -0.270
ResNet-50	Unweighted BCE	Pleural Effusion	32.18%	0.2182	0.677 [0.645, 0.709]	0.2430	–	-0.113
ResNet-50	Unweighted BCE	Cardiomegaly	26.33%	0.1940	0.519 [0.489, 0.548]	0.2333	–	-0.203
ResNet-50	Unweighted BCE	Pneumonia	10.05%	0.0904	0.583 [0.540, 0.623]	0.0982	–	-0.086
ResNet-50	Unweighted BCE	Edema	21.33%	0.1678	0.516 [0.480, 0.552]	0.2126	–	-0.266
ResNet-50	Pos-Weighted BCE	Pleural Effusion	32.18%	0.2182	0.620 [0.588, 0.653]	0.2665 $\rightarrow$ 0.2557	-0.0109 [-0.0417, +0.0253]	-0.221 $\rightarrow$ -0.172
ResNet-50	Pos-Weighted BCE	Cardiomegaly	26.33%	0.1940	0.522 [0.491, 0.552]	0.2822 $\rightarrow$ 0.2211	-0.0611 [-0.0818, -0.0415]	-0.455 $\rightarrow$ -0.140
ResNet-50	Pos-Weighted BCE	Pneumonia	10.05%	0.0904	0.589 [0.545, 0.629]	0.1137 $\rightarrow$ 0.0982	-0.0155 [-0.0254, -0.0059]	-0.257 $\rightarrow$ -0.086
ResNet-50	Pos-Weighted BCE	Edema	21.33%	0.1678	0.570 [0.538, 0.602]	0.1929 $\rightarrow$ 0.2130	+0.0201 [+0.0092, +0.0313]	-0.149 $\rightarrow$ -0.269

Section S3: Diagnostic Study-Level Sensitivity Analysis

To evaluate sensitivity to repeated within-patient examinations and multi-image studies, Table S2 summarizes external validation performance aggregated strictly at the diagnostic study level ($N = 3,041$ studies across $N = 289$ unique patients). In accordance with standard radiological diagnostic practice, multiple views acquired during a single examination are aggregated using maximum probability pooling ($\max_i p_i$, where detection on any view indicates study-level positive status). Supplementary Table S2 evaluates all 16 architecture-loss-pathology combinations with patient-clustered 1,000-replicate bootstrap 95% confidence intervals. As shown, study-level results demonstrate that the principal findings were qualitatively concordant regardless of whether the statistical unit is the individual radiograph or the diagnostic study: unweighted models yield comparable discrimination across views, while external discrimination remains pathology-dependent (AUROC approaching chance for Cardiomegaly), and analytic logit recalibration ($z - \log w$) on positive-weighted models yields statistically significant reductions in study-level Brier error for Cardiomegaly ($\Delta\text{Brier} = -0.0601$ on DenseNet-121 and -0.0642 on ResNet-50) and Pneumonia ($\Delta\text{Brier} = -0.0100$ and -0.0163 , both intervals strictly below zero), alongside consistent point-estimate reductions for Pleural Effusion (-0.0102 and -0.0118).

Table S2: Supplementary Table S2: Diagnostic Study-Level Sensitivity Analysis on the Official MIMIC-CXR-JPG Test Cohort ($N = 3,041$ Studies across $N = 289$ Unique Patients). Multi-image studies are aggregated using maximum probability pooling ($\max_i p_i$). Evaluated on 3-seed probability ensembles with patient-clustered 1,000-replicate bootstrap 95% confidence intervals, local null Brier reference ($\text{Brier}_{\text{null}} = \pi(1 - \pi)$), and paired recalibration contrast ($\Delta\text{Brier} = \text{Brier}_{\text{corr}} - \text{Brier}_{\text{raw}}$).

Architecture	Objective	Pathology	Study Prevalence	Study $\text{Brier}_{\text{null}}$	AUROC [95% CI]	Brier (Raw $\rightarrow$ Corr)	ΔBrier [95% CI]	BSS (Raw $\rightarrow$ Corr)
DenseNet-121	Unweighted BCE	Pleural Effusion	32.55%	0.2196	0.695 [0.663, 0.723]	0.2488	–	-0.133
DenseNet-121	Unweighted BCE	Cardiomegaly	26.54%	0.1951	0.526 [0.497, 0.559]	0.2378	–	-0.219
DenseNet-121	Unweighted BCE	Pneumonia	10.16%	0.0913	0.608 [0.562, 0.650]	0.0996	–	-0.091
DenseNet-121	Unweighted BCE	Edema	21.64%	0.1697	0.565 [0.528, 0.600]	0.2157	–	-0.271
DenseNet-121	Pos-Weighted BCE	Pleural Effusion	32.55%	0.2196	0.674 [0.643, 0.704]	0.2556 $\rightarrow$ 0.2453	-0.0102 [-0.0427, +0.0236]	-0.164 $\rightarrow$ -0.117
DenseNet-121	Pos-Weighted BCE	Cardiomegaly	26.54%	0.1951	0.527 [0.497, 0.558]	0.2933 $\rightarrow$ 0.2332	-0.0601 [-0.0814, -0.0367]	-0.503 $\rightarrow$ -0.195
DenseNet-121	Pos-Weighted BCE	Pneumonia	10.16%	0.0913	0.599 [0.555, 0.639]	0.1091 $\rightarrow$ 0.0992	-0.0100 [-0.0179, -0.0018]	-0.195 $\rightarrow$ -0.086
DenseNet-121	Pos-Weighted BCE	Edema	21.64%	0.1697	0.583 [0.550, 0.616]	0.1937 $\rightarrow$ 0.2165	+0.0228 [+0.0126, +0.0325]	-0.141 $\rightarrow$ -0.275
ResNet-50	Unweighted BCE	Pleural Effusion	32.55%	0.2196	0.682 [0.650, 0.714]	0.2431	–	-0.107
ResNet-50	Unweighted BCE	Cardiomegaly	26.54%	0.1951	0.521 [0.494, 0.548]	0.2340	–	-0.200
ResNet-50	Unweighted BCE	Pneumonia	10.16%	0.0913	0.584 [0.543, 0.624]	0.0992	–	-0.087
ResNet-50	Unweighted BCE	Edema	21.64%	0.1697	0.516 [0.480, 0.553]	0.2158	–	-0.272
ResNet-50	Pos-Weighted BCE	Pleural Effusion	32.55%	0.2196	0.625 [0.594, 0.655]	0.2685 $\rightarrow$ 0.2567	-0.0118 [-0.0427, +0.0251]	-0.223 $\rightarrow$ -0.169
ResNet-50	Pos-Weighted BCE	Cardiomegaly	26.54%	0.1951	0.523 [0.493, 0.552]	0.2858 $\rightarrow$ 0.2217	-0.0642 [-0.0852, -0.0434]	-0.465 $\rightarrow$ -0.136
ResNet-50	Pos-Weighted BCE	Pneumonia	10.16%	0.0913	0.588 [0.542, 0.627]	0.1155 $\rightarrow$ 0.0992	-0.0163 [-0.0267, -0.0068]	-0.265 $\rightarrow$ -0.086
ResNet-50	Pos-Weighted BCE	Edema	21.64%	0.1697	0.571 [0.538, 0.606]	0.1951 $\rightarrow$ 0.2164	+0.0213 [+0.0100, +0.0330]	-0.149 $\rightarrow$ -0.275

Section S4: Complete In-Domain Factorial Evaluation Matrix

Table S3 provides the comprehensive 17-column longitudinal evaluation matrix across all models, loss objectives, and thoracic abnormalities across deidentified-date strata T_1 , T_2 , and T_3 , displaying discrimination (AUROC), calibration accuracy (Brier score, ECE), and negative log-likelihood (log-loss) alongside mean and inter-seed standard deviations across random seeds $S \in \{42, 123, 2026\}$.

Table S3: Supplementary Table S3: Complete 17-Column Longitudinal Factorial Matrix across DenseNet-121 and ResNet-50 Architectures, Loss Objectives, and Target Thoracic Pathologies across Strata T_1 (2015-H2), T_2 (2016), and T_3 (2017). Values represent mean $\pm$ standard deviation across three random seeds.

Architecture	Objective	Target	AUROC T_1	AUROC T_2	AUROC T_3	AUROC Δ	Brier T_1	Brier T_2	Brier T_3	Brier Δ	ECE T_1	ECE T_2	ECE Δ	LogLoss T_1	LogLoss T_2	LogLoss Δ
DenseNet-121	Unweighted BCE	Pleural Effusion	0.9335 $\pm$ 0.0052	0.8895 $\pm$ 0.0115	0.8731 $\pm$ 0.0120	-0.0604	0.0369 $\pm$ 0.0029	0.0356 $\pm$ 0.0009	0.0279 $\pm$ 0.0009	-0.0090	0.0232 $\pm$ 0.0078	0.0090 $\pm$ 0.0022	-0.0143	0.1318 $\pm$ 0.0076	0.1114 $\pm$ 0.0027	-0.0204
DenseNet-121	Unweighted BCE	Cardiomegaly	0.9075 $\pm$ 0.0116	0.8965 $\pm$ 0.0140	0.9017 $\pm$ 0.0170	-0.0057	0.0532 $\pm$ 0.0012	0.0677 $\pm$ 0.0021	0.0677 $\pm$ 0.0028	+0.0145	0.0140 $\pm$ 0.0009	0.0210 $\pm$ 0.0074	+0.0071	0.1798 $\pm$ 0.0072	0.2231 $\pm$ 0.0102	+0.0433
DenseNet-121	Unweighted BCE	Pneumonia	0.8473 $\pm$ 0.0129	0.8818 $\pm$ 0.0166	0.8647 $\pm$ 0.0172	+0.0174	0.0235 $\pm$ 0.0008	0.0182 $\pm$ 0.0003	0.0187 $\pm$ 0.0006	-0.0048	0.0142 $\pm$ 0.0011	0.0095 $\pm$ 0.0016	-0.0046	0.0996 $\pm$ 0.0035	0.0898 $\pm$ 0.0039	-0.0187
DenseNet-121	Unweighted BCE	Edema*	0.5402 $\pm$ 0.0467	0.8640 $\pm$ 0.0886	-	-	0.0021 $\pm$ 0.0000	0.0032 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	-0.0021	0.0029 $\pm$ 0.0013	0.0029 $\pm$ 0.0029	-0.0000	0.0063 $\pm$ 0.0008	0.0029 $\pm$ 0.0029	-0.0134
DenseNet-121	Weighted BCE	Pleural Effusion	0.9061 $\pm$ 0.0537	0.8679 $\pm$ 0.0457	0.8391 $\pm$ 0.0657	-0.0670	0.0881 $\pm$ 0.0394	0.1287 $\pm$ 0.0432	0.1254 $\pm$ 0.0456	+0.0173	0.1964 $\pm$ 0.0702	0.2227 $\pm$ 0.0705	+0.0264	0.3646 $\pm$ 0.1179	0.4100 $\pm$ 0.1279	+0.0454
DenseNet-121	Weighted BCE	Cardiomegaly	0.8749 $\pm$ 0.0540	0.8599 $\pm$ 0.0600	0.8575 $\pm$ 0.0609	-0.0164	0.1262 $\pm$ 0.0247	0.1558 $\pm$ 0.0316	0.1300 $\pm$ 0.0306	-0.0098	0.2031 $\pm$ 0.0507	0.1969 $\pm$ 0.0486	-0.0062	0.2892 $\pm$ 0.0807	0.4241 $\pm$ 0.0906	+0.0259
DenseNet-121	Weighted BCE	Pneumonia	0.8191 $\pm$ 0.0296	0.8494 $\pm$ 0.0313	0.8424 $\pm$ 0.0496	+0.0163	0.1313 $\pm$ 0.0533	0.1308 $\pm$ 0.0555	0.1298 $\pm$ 0.0523	-0.0015	0.2393 $\pm$ 0.0894	0.2425 $\pm$ 0.0883	+0.0032	0.4226 $\pm$ 0.1511	0.4166 $\pm$ 0.1471	-0.0060
DenseNet-121	Weighted BCE	Edema*	0.6122 $\pm$ 0.0268	0.8753 $\pm$ 0.0129	-	-	0.0614 $\pm$ 0.0637	0.0672 $\pm$ 0.0703	0.0638 $\pm$ 0.0734	+0.0022	0.1191 $\pm$ 0.1228	0.1274 $\pm$ 0.1327	+0.0082	0.2095 $\pm$ 0.2021	0.2093 $\pm$ 0.2288	-0.0002
ResNet-50	Unweighted BCE	Pleural Effusion	0.9296 $\pm$ 0.0113	0.8755 $\pm$ 0.0203	0.8608 $\pm$ 0.0190	-0.0687	0.0372 $\pm$ 0.0022	0.0364 $\pm$ 0.0013	0.0284 $\pm$ 0.0010	-0.0088	0.0162 $\pm$ 0.0050	0.0132 $\pm$ 0.0030	-0.0029	0.1294 $\pm$ 0.0099	0.1145 $\pm$ 0.0067	-0.0149
ResNet-50	Unweighted BCE	Cardiomegaly	0.8808 $\pm$ 0.0177	0.8756 $\pm$ 0.0317	0.8785 $\pm$ 0.0196	-0.0112	0.0572 $\pm$ 0.0037	0.0709 $\pm$ 0.0050	0.0712 $\pm$ 0.0051	+0.0140	0.0233 $\pm$ 0.0100	0.0175 $\pm$ 0.0081	-0.0058	0.1954 $\pm$ 0.0103	0.2379 $\pm$ 0.0174	+0.0425
ResNet-50	Unweighted BCE	Pneumonia	0.8432 $\pm$ 0.0178	0.8629 $\pm$ 0.0130	0.8798 $\pm$ 0.0169	+0.0366	0.0242 $\pm$ 0.0007	0.0187 $\pm$ 0.0003	0.0191 $\pm$ 0.0003	-0.0051	0.0128 $\pm$ 0.0050	0.0106 $\pm$ 0.0015	-0.0023	0.1024 $\pm$ 0.0059	0.0899 $\pm$ 0.0035	-0.0215
ResNet-50	Unweighted BCE	Edema*	0.5564 $\pm$ 0.0368	0.8966 $\pm$ 0.0216	-	-	0.0021 $\pm$ 0.0000	0.0032 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	-0.0021	0.0023 $\pm$ 0.0005	0.0017 $\pm$ 0.0009	-0.0006	0.0175 $\pm$ 0.0010	0.0017 $\pm$ 0.0009	-0.0158
ResNet-50	Weighted BCE	Pleural Effusion	0.8790 $\pm$ 0.0508	0.8467 $\pm$ 0.0458	0.8191 $\pm$ 0.0496	-0.0598	0.1256 $\pm$ 0.0325	0.1437 $\pm$ 0.0277	0.1392 $\pm$ 0.0310	+0.0137	0.2481 $\pm$ 0.0731	0.2751 $\pm$ 0.0709	+0.0270	0.4803 $\pm$ 0.0895	0.4465 $\pm$ 0.0750	+0.0361
ResNet-50	Weighted BCE	Cardiomegaly	0.8298 $\pm$ 0.0550	0.8046 $\pm$ 0.0701	0.7990 $\pm$ 0.0788	-0.0308	0.1415 $\pm$ 0.0124	0.1526 $\pm$ 0.0083	0.1322 $\pm$ 0.0085	+0.0107	0.2450 $\pm$ 0.0348	0.2338 $\pm$ 0.0258	-0.0112	0.4427 $\pm$ 0.0380	0.4678 $\pm$ 0.0305	+0.0251
ResNet-50	Weighted BCE	Pneumonia	0.8189 $\pm$ 0.0199	0.8359 $\pm$ 0.0345	0.8233 $\pm$ 0.0533	-0.0116	0.1667 $\pm$ 0.0584	0.1681 $\pm$ 0.0583	0.1693 $\pm$ 0.0613	+0.0024	0.3189 $\pm$ 0.1175	0.3253 $\pm$ 0.1208	+0.0064	0.5696 $\pm$ 0.1443	0.5096 $\pm$ 0.1327	+0.0600
ResNet-50	Weighted BCE	Edema*	0.5490 $\pm$ 0.0852	0.8310 $\pm$ 0.0437	-	-	0.0093 $\pm$ 0.0461	0.0749 $\pm$ 0.0509	0.0748 $\pm$ 0.0494	+0.0052	0.1545 $\pm$ 0.1121	0.1670 $\pm$ 0.1169	+0.0126	0.2193 $\pm$ 0.1374	0.2255 $\pm$ 0.1469	+0.0026

Section S5: Complete Multi-Calibrator Parameter Matrix

Table S4 documents the exhaustive 12-column multi-calibrator evaluation on Pleural Effusion across 3-seed probability ensembles on held-out deidentified-date strata T_1 , T_2 , and T_3 , including Cox calibration regression intercepts (α), calibration slopes (β), ECE, log-loss, and patient-clustered 2,000-replicate bootstrap contrasts against uncalibrated models.

Table S4: Supplementary Table S4: Complete 12-Column Multi-Calibrator Comparison on Pleural Effusion across 3-Seed Probability Ensembles, Evaluating Cox Intercept (α), Slope (β), ECE, Log-Loss, and Paired Bootstrap Contrasts.

Architecture	Loss Objective	Calibrator	T1 Brier	T2 Brier	T3 Brier	Δ Brier (T3 - T1)	T3 Intercept (α)	T3 Slope (β)	T3 ECE	T3 Log-Loss	T3 Δ Brier vs Raw	95% Bootstrap CI]
DenseNet-121	Weighted BCE	Raw (Uncalibrated)	0.0920	0.1111	0.1079	+0.0158	-2.9638	1.0484	0.2227	0.3572	-	-
DenseNet-121	Weighted BCE	Temperature Scaling (T*)	0.0911	0.1100	0.1067	+0.0157	-2.9615	1.0289	0.2190	0.3540	-0.0011	[-0.0014, -0.0009]
DenseNet-121	Weighted BCE	Analytic Offset ($z - \log w$)	0.0359	0.0348	0.0284	-0.0075	-0.4451	0.9292	0.0126	0.1134	-0.0795	[-0.0893, -0.0697]
DenseNet-121	Weighted BCE	Fitted Offset ($z + a^*$)	0.0354	0.0352	0.0289	-0.0066	-0.6013	0.9423	0.0183	0.1148	-0.0790	[-0.0885, -0.0696]
DenseNet-121	Weighted BCE	Platt Scaling ($a + bz$)	0.0357	0.0345	0.0281	-0.0076	-0.2288	1.1280	0.0176	0.1136	-0.0798	[-0.0894, -0.0702]
DenseNet-121	Unweighted BCE	Raw (Uncalibrated)	0.0356	0.0341	0.0268	-0.0088	-0.1471	1.0439	0.0040	0.1054	-	-
DenseNet-121	Unweighted BCE	Temperature Scaling (T*)	0.0350	0.0339	0.0268	-0.0082	+0.1528	1.1735	0.0082	0.1066	+0.0001	[-0.0003, +0.0004]
DenseNet-121	Unweighted BCE	Fitted Offset ($z + a^*$)	0.0333	0.0343	0.0274	-0.0059	-0.4177	1.0422	0.0160	0.1089	+0.0007	[-0.0006, +0.0018]
DenseNet-121	Unweighted BCE	Platt Scaling ($a + bz$)	0.0333	0.0355	0.0285	-0.0048	-0.7231	0.8828	0.0175	0.1102	+0.0017	[-0.0000, +0.0034]
ResNet-50	Weighted BCE	Raw (Uncalibrated)	0.1101	0.1254	0.1227	+0.0126	-2.8526	1.9498	0.2751	0.4015	-	-
ResNet-50	Weighted BCE	Temperature Scaling (T*)	0.0876	0.1045	0.1008	+0.0131	-2.6080	1.1840	0.1993	0.3187	-0.0220	[-0.0241, -0.0198]
ResNet-50	Weighted BCE	Analytic Offset ($z - \log w$)	0.0402	0.0367	0.0291	-0.0111	+1.0751	1.4052	0.0138	0.1149	-0.0936	[-0.1022, -0.0844]
ResNet-50	Weighted BCE	Fitted Offset ($z + a^*$)	0.0384	0.0361	0.0291	-0.0093	+0.7008	1.5504	0.0229	0.1208	-0.0936	[-0.1016, -0.0851]
ResNet-50	Weighted BCE	Platt Scaling ($a + bz$)	0.0380	0.0364	0.0297	-0.0082	-0.1911	1.1777	0.0223	0.1211	-0.0930	[-0.1007, -0.0849]
ResNet-50	Unweighted BCE	Raw (Uncalibrated)	0.0361	0.0351	0.0272	-0.0088	-0.0766	1.0945	0.0109	0.1087	-	-
ResNet-50	Unweighted BCE	Temperature Scaling (T*)	0.0361	0.0351	0.0273	-0.0088	-0.0957	1.0974	0.0107	0.1089	+0.0000	[-0.0001, +0.0001]
ResNet-50	Unweighted BCE	Fitted Offset ($z + a^*$)	0.0352	0.0354	0.0277	-0.0074	-0.3631	1.0755	0.0189	0.1113	+0.0005	[-0.0001, +0.0011]
ResNet-50	Unweighted BCE	Platt Scaling ($a + bz$)	0.0351	0.0363	0.0284	-0.0066	-0.7099	0.8891	0.0204	0.1124	+0.0012	[-0.0001, +0.0024]

Section S6: Complete Selective Prediction Workload Triage Matrix

Table S5 provides the complete 24-row selective prediction and human-in-the-loop referral simulation on later held-out stratum T_3 ($N = 2,436$ radiographs, 85 true positive Pleural Effusion cases) across coverage deciles from 100% down to 50%, detailing automated decisions, referred cases, retained true and false positives, retained sensitivity, full-cohort system triage sensitivity, retained AUROC, and retained Brier score.

Table S5: Supplementary Table S5: Complete 24-Row Selective Prediction and Reader Referral Triage Simulation on Later Held-Out Stratum T_3 ($N = 2,436$ Radiographs, 85 Positive Pleural Effusion Cases) across All Coverage Regimes (100% to 50%).

Coverage	Strategy	Automated Decisions (N)	Referred to Reader (N)	Retained Positives (N)	Automated True Positives (N)	Automated False Negatives (Missed Cases)	Retained Sensitivity	Full-Cohort Triage Sensitivity	Retained AUROC	Retained Brier
100%	Weighted BCE (Raw)	2,436	0 (0%)	85	69	16.0	0.8118	0.8118	0.8691	0.1079
100%	Weighted BCE (Analytic Corrected)	2,436	0 (0%)	85	69	16.0	0.8118	0.8118	0.8693	0.1084
100%	Unweighted BCE (Raw)	2,436	0 (0%)	85	73	12.0	0.8588	0.8588	0.8806	0.1068
100%	Random Deferral Baseline	2,436	0 (0%)	85.0	69.0	16.0	0.8118	0.8118	0.8691	0.1079
90%	Weighted BCE (Raw)	2,193	243 (9%)	82	68	14.0	0.8293	0.8293	0.8789	0.1062
90%	Weighted BCE (Analytic Corrected)	2,193	243 (9%)	79	64	15.0	0.8101	0.8235	0.8735	0.1029
90%	Unweighted BCE (Raw)	2,193	243 (9%)	78	66	12.0	0.8588	0.8662	0.8923	0.1056
90%	Random Deferral Baseline	2,193	243 (9%)	76.5	62.0	14.5	0.8106	0.8295	0.8825	0.1080
80%	Weighted BCE (Raw)	1,949	487 (19%)	77	67	10.0	0.8701	0.8824	0.8882	0.1012
80%	Weighted BCE (Analytic Corrected)	1,949	487 (19%)	67	52	15.0	0.7701	0.8235	0.8896	0.1024
80%	Unweighted BCE (Raw)	1,949	487 (19%)	69	57	12.0	0.8261	0.8588	0.9027	0.1026
80%	Random Deferral Baseline	1,949	487 (19%)	68.0	55.0	13.0	0.8090	0.8472	0.9007	0.1080
70%	Weighted BCE (Raw)	1,706	730 (30%)	66	59	7.0	0.8939	0.9176	0.8997	0.0956
70%	Weighted BCE (Analytic Corrected)	1,706	730 (30%)	44	31	13.0	0.7045	0.8471	0.8876	0.1178
70%	Unweighted BCE (Raw)	1,706	730 (30%)	50	39	11.0	0.7800	0.8706	0.9051	0.1034
70%	Random Deferral Baseline	1,706	730 (30%)	59.5	48.2	11.3	0.8099	0.9069	0.9069	0.1081
60%	Weighted BCE (Raw)	1,462	974 (40%)	36	31	5.0	0.9107	0.9412	0.9058	0.0866
60%	Weighted BCE (Analytic Corrected)	1,462	974 (40%)	20	11	9.0	0.5500	0.8941	0.8176	0.0905
60%	Unweighted BCE (Raw)	1,462	974 (40%)	29	22	7.0	0.7586	0.9176	0.8788	0.1020
60%	Random Deferral Baseline	1,462	974 (40%)	31.0	41.4	9.6	0.8122	0.8873	0.9180	0.1080
50%	Weighted BCE (Raw)	1,218	1,218 (50%)	30	45	5.0	0.9000	0.9412	0.9003	0.0834
50%	Weighted BCE (Analytic Corrected)	1,218	1,218 (50%)	7	2	5.0	0.2657	0.9412	0.5572	0.0551
50%	Unweighted BCE (Raw)	1,218	1,218 (50%)	14	7	7.0	0.9000	0.9176	0.8054	0.0880
50%	Random Deferral Baseline	1,218	1,218 (50%)	42.5	34.4	8.1	0.8101	0.9051	0.9051	0.1078

Section S7: Reproducibility and Code Execution Guide

The complete codebase is organized to support deterministic replication from raw prediction archives through all tables, figures, and statistical tests:

1. **Environment Setup:** Python 3.10+ with PyTorch, torchvision, scikit-learn, scipy, pandas, numpy, and matplotlib.
2. **Stage 3 Recalibration Execution:**

```
python3 scripts/run_stage3_recalibration.py
```

Computes probability ensembles ($\bar{p}_m = \frac{1}{3} \sum_{s=1}^3 p_{m,s}$) and ensemble logit offsets ($\Delta z = \frac{1}{3} \sum_{s=1}^3 (z_{w,s} - z_{u,s})$) across all 8,302 held-out radiographs (16,604 paired image-architecture evaluations, alongside all 49,812 seed-specific pairs) for raw, temperature-scaled, analytic offset ($z - \log w$), fitted offset ($z + a^*$), and Platt-scaled models, running 2,000-replicate patient-clustered bootstrap inference.

3. **Publication Artifact Compilation:**

```
python3 scripts/compile_publication_artifacts.py
```

Generates Tables 1–4 in LaTeX and markdown formats, alongside publication figures (Figures 2–6).

4. **Verification Audit:**

```
python3 scripts/audit_cmpb.py
```

Verifies all formatting, word count, highlight length (≤ 85 characters), and statutory compliance rules for Computer Methods and Programs in Biomedicine.